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1. Diagonal matrices in MATLAB

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Recitation due Feb 14, 2024 20:30 IST

Diagonal matrices (External resource) (1.0 points possible)

Diagonal matrices

MATLAB's command `diag()` has two uses

- If $\mathbf{v}$ is a vector of length n , then

`D = diag(v);`
creates an $n \times n$ diagonal matrix with the elements of $\mathbf{v}$ along the diagonal

- If $\mathbf{A}$ is an $n \times n$ square matrix, then

`x = diag(A)`
creates a column vector $\mathbf{x}$ out of the elements from the main diagonal of $\mathbf{A}$.

Complete the script below, which illustrates the two uses of the `diag` command.

Script ?

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 MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 % Firstly create a random 15x15 matrix A using the rand() command
2 A = rand(15,15);
3 % Now create a new variable v which is a column vector
4 % made from the main diagonal of A using the diag() function
5 v = diag(A);
6 % Now create a 10x10 diagonal matrix D made from the first ten elements of v
7 % Think carefully about the easiest way to do this. Remember you can create a vector of zeros.
8 D = diag(v(1:10));
```

 Run Script ?

Upper triangular matrices (External resource)

(1.0 points possible)

Upper triangular matrices

 Calculator

Given any square matrix A , MATLAB can create an upper triangular matrix X out of the upper triangular elements of A using the following command

```
X = triu(A)
```

In the MATLAB script below, we demonstrate that the determinant of an upper triangular matrix is the product of its diagonal elements. Fill in the missing pieces in the script, and run the script a couple of times for yourself to see that the result holds!

Script ?

 Save

 Reset

 [MATLAB Documentation \(https://www.mathworks.com/help/\)](https://www.mathworks.com/help/)

```
1 % Firstly generate a random 10X10 matrix A using the rand() command
2 A = rand(10);
3 % Now define an upper triangular matrix X made out of the upper triangular elements of A
4 X = triu(A);
5 % Calculate the determinant of X explicitly using det(). Define the variable d1
6 d1 = det(X);
7 % Finally extract the main diagonal of X as a column vector v by using diag()
8 % We then calculate the product of the elements of v and call this d2.
9 v = diag(X);
10 d2 = prod(v)
11 a = d1-d2;
12
```

 Run Script



Assessment: All Tests Passed

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-  **Determinant is product of diagonal entries**
-  **A correctly defined**

1. Diagonal matrices in MATLAB

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